INNOVATEX HACKATHON 2024

YAR TECH SERVICES

Problem Statement :AI-Enhanced Deepfake Detection Tool for Social Media

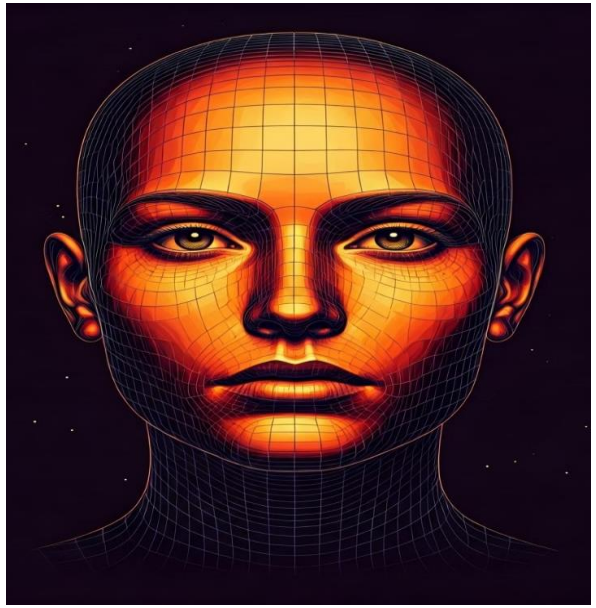
Domain: ARTIFICIAL INTELLIGENCE

TEAM NAME: SciPy

IDEA TITLE

Introduction:

•Deepfake videos are synthetic media generated using AI techniques like **Generative Adversarial Networks (GANs)** and **autoencoders**.



•**How Deepfakes are Created**:

•Voice synthesis

•Lip-syncing

•**Detection Challenge**: Deepfakes can mimic real content with high accuracy, making detection difficult.

•Face swapping

IDEA/Solution:

•Data sources like FaceForensics++ and Celeb-DF are used to train models on real and fake videos.

•Deep learning models, such as CNNs and transfer learning with pre-trained architectures like Xception, help classify deepfakes from real content.

• BYUsing feature extraction (e.g., facial landmarks), model training, and real-time inference.

•The system can be deployed with a user interface for real-time video classification, ensuring the authenticity of online media.

TECHNICAL APPROACH

- Tools and Libraries:

1. **Python** (Programming Language)
2. **OpenCV**(For Video Processing)
3. **TensorFlow/PyTorch**(Deep Learning Frameworks)
4. **MediaPipe**(For Facial Landmark Detection)
5. **FFmpeg** (For Video Frame Extraction)
6. **NumPy and Matplotlib** (For Data Manipulation and Visualization)
7. **Scikit-Learn** (For Machine Learning Algorithms)

FEASIBILITY AND VIABILITY

- FEASIBILITY:

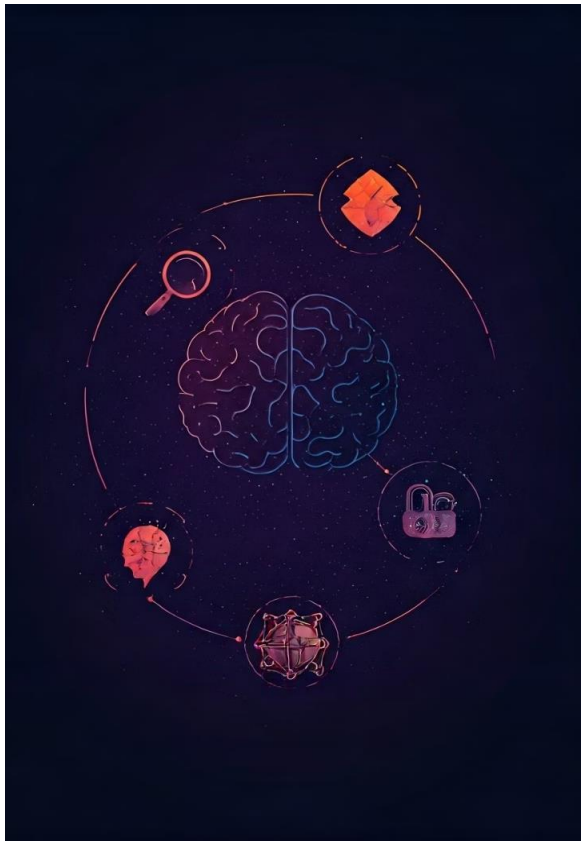
  Technical Feasibility:Deep Learning Frameworks,
  Pre-trained Models, Transfer Learning, Face Detection Tools.
  Financial Feasibility:Initial Investment for hardware software and data costs.

  Monetization Potential License to Enterprises, Real-time Deepfake Detection,

  Maintenance Costs

- VIABILITY:

  Market Viability
  Demand for Deepfake Detection
  Competition



Research Papers and Academic References

"DeepFake Detection: A Survey"

- **Authors**: R. Dolhansky, A. G. DePuy, A. Jain, and R. T. Edwards

"Exploiting DeepFake Video Detection Methods for Real-World Application"

- **Authors**: A. Rossler, D. Cozzolino, and F. Iuliani

"DeepFace: Closing the Gap to Human-Level Performance in Face Verification"

- **Authors**: Y. Taigman, M. Yang, M. Ranzato, and L. Wolf

Datasets for Deepfake Detection

FaceForensics++

Celeb-DF

DFDC (DeepFake Detection Challenge)